Information Security : Access Control

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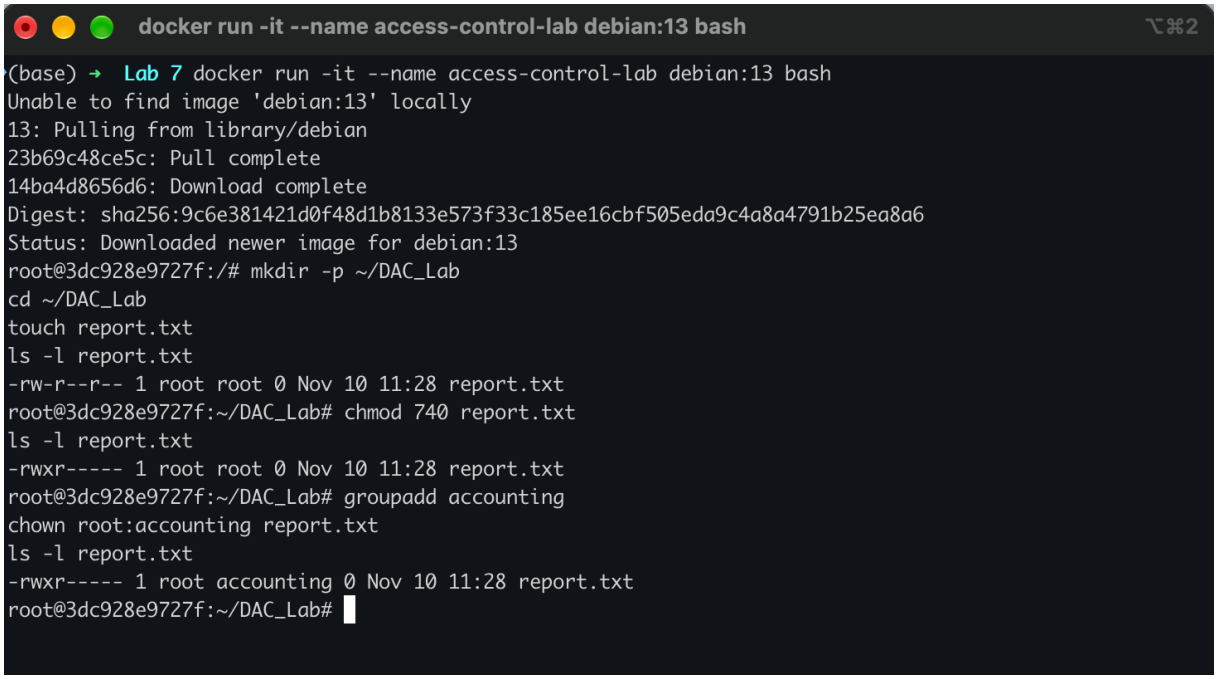
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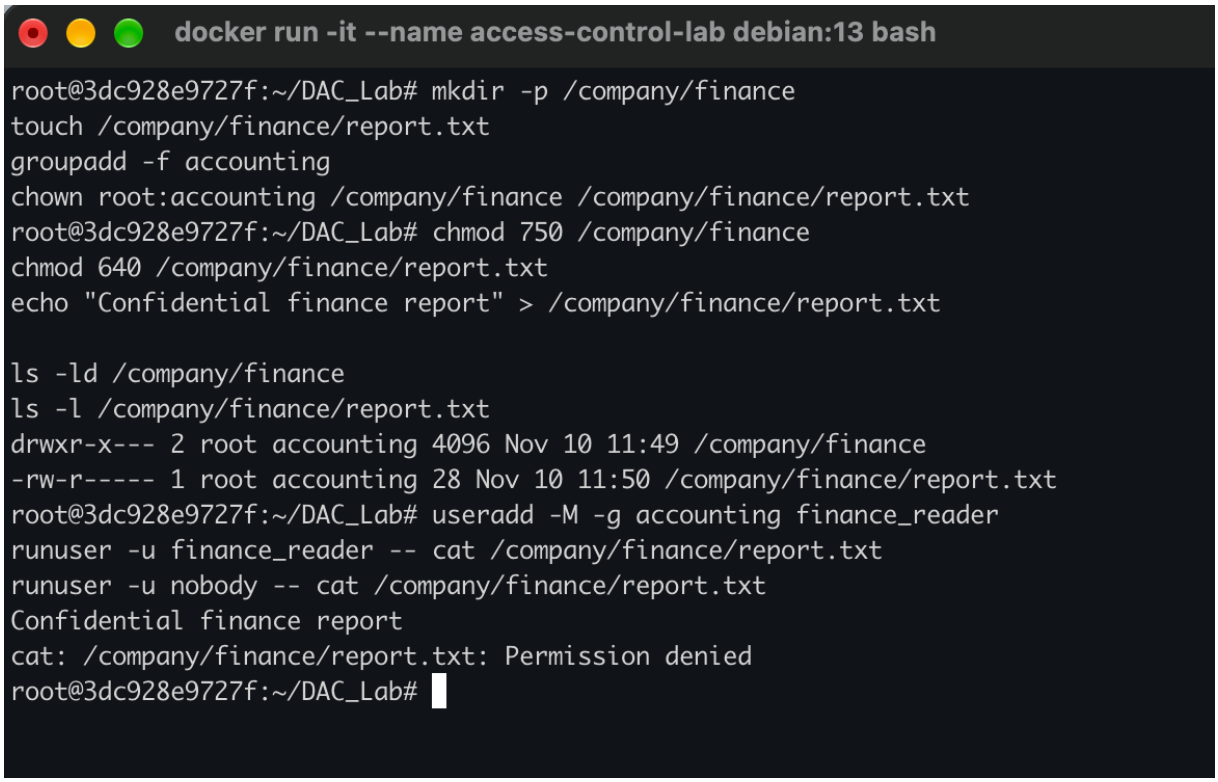
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1 DAC

1.1 Warmup



1.2 Challenge 1



6. No. Mode `750` lets accounting members traverse the directory, but mode `600` gives read permission only to the file's owner.

2 MAC

2.1 Challenge 2

1. Read access

User	A (Secret)	B (Confidential)	C (Public)
Alice (Secret)	Yes	Yes	Yes
Bob (Confidential)	No	Yes	Yes
Charlie (Public)	No	No	Yes

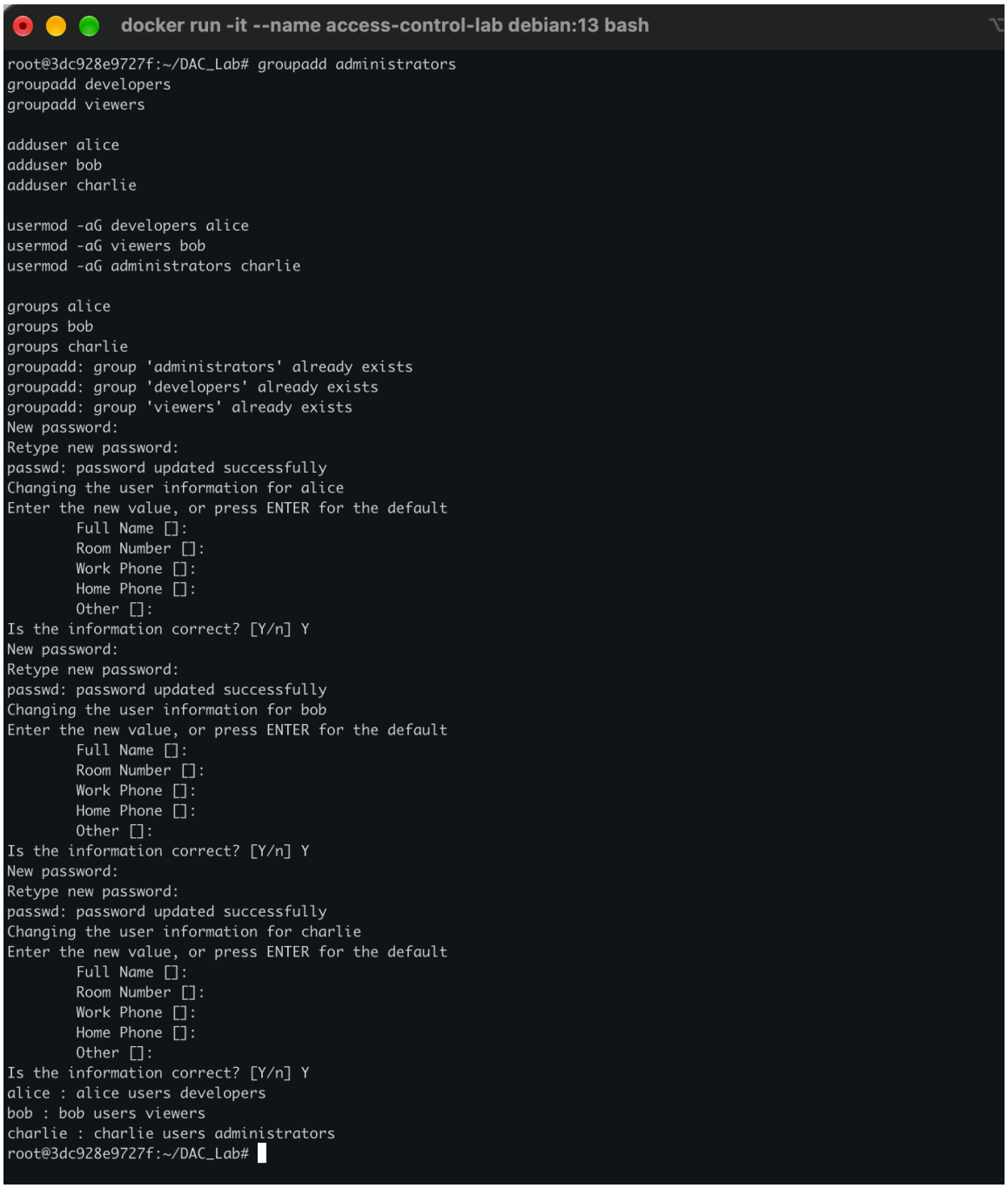
2. Denied. Document A is above Bob's clearance (no read up).

3. Denied. Alice is operating at Secret, and Document C is Public (no write down).

4. No read up blocks lower-level users from reading classified data. No write down blocks higher-level users from copying it into lower-level documents.

5. No. DAC checks ordinary file permissions set by the owner, while MAC checks the central policy. Both checks must allow access.

2.2 Challenge 3



3. No. Developers could not read `/company/reports`, and administrators had no access to either directory.



3 ABAC

3.1 Challenge 4

1-2. Access decisions

User	Decision	Reason
Alice	Granted	All three conditions are met.
Bob	Denied	His department is HR and the document is Finance.
Charlie	Denied	His Public clearance is below Secret.
David	Denied	22:00 is outside the allowed hours.

3. Denied. Alice works in Finance and the document belongs to HR.
4. Granted. At 17:00 all three conditions are met.
5. ABAC can combine user, document, and context attributes, so the same user can receive different decisions depending on the document or time.